

4.12 Writing Equations of Parallel and Perpendicular Lines

Lesson Introduction

 Benchmark	MA.912.AR.2.3 Write a linear two-variable equation for a line that is parallel or perpendicular to a given line and goes through a given point.		 Learning Targets	- I can write an equation for a line parallel to a given line through a given point. - I can write an equation for a line perpendicular to a given line through a given point.
 Benchmark Coherence	Building On MA.8.AR.4.3 Given a mathematical or real-world context, solve systems of two linear equations by graphing.		Working Toward MA.912.GR.3.2 Given a mathematical or real-world context, use coordinate geometry to classify or justify definitions, properties and theorems involving circles, triangles, or quadrilaterals.	
 Essential Question	How does a line being parallel or perpendicular to another line inform your approach to writing its equation?			
 Lesson Narrative	Lesson 12 serves as the culminating application for Lessons 9-11. Students have explored, formalized, and applied skills and concepts of parallel and perpendicular lines and their equations, including using contexts of horizontal and vertical lines. This lesson synthesizes their findings to apply what they have learned in a variety of contexts and helps make decisions for when and how to apply their learnings.			
 Component Summary	4.12.1 Warm-Up (~ 5 min.)	Spiraled practice for writing the equation of lines (<i>SE p. 225</i>)	4.12.3 Your Turn! (10 min.)	Independent practice for students (<i>SE p. 226</i>)
	4.12.2 Collaboration (25 min.)	Card Sort collaborative activity for synthesizing and applying learning from Lessons 9-11 (<i>SE p. 225</i>)	4.12.4 Wrap-Up (~10 min.)	Error analysis in writing the equations of perpendicular lines (<i>SE p. 228</i>)
 Resources	Glossary		Materials	
	<u>Key Terms:</u> N/A <u>Additional Glossary:</u> - Parallel - Perpendicular - Reciprocal		Card Sort	
			Additional Preparation N/A	

 Teacher Tips	Common Misconceptions	Things to Consider
	Students may struggle with conceptualizing the difference between a slope of zero and an undefined slope, leading to an incorrect response in 4.12.3 Your Turn! question 3.	Utilize the resources, data, and informal observations from Lessons 9-11 to inform intentional groups during this lesson. That could mean teacher-led small groups based on data from Lessons 9-11, or specific remediation strategies for students during 4.12.3 Your Turn! based on previous lessons.
	Literacy and Language Routines	Mathematical Thinking and Reasoning (MTR) Standard
	Take Turns 4.12.2 Collaboration allows students to engage in a collaborative activity on three features of lines. Give students the opportunity to try each problem in the activity, then do an agree/disagree/justify opportunity. The “Take Turns” framing helps students monitor airtime and provides equitable opportunity for engagement.	MA.K12.MTR.2.1: Multiple Representations Utilize the Card Sort in 4.12.2 Collaboration to provide opportunity for students to practice and engage in meaningful math thinking and reasoning with multiple representations. Highlight student work that exemplifies the clarification language within this standard. “connections between concepts and representations...” “concrete to pictorial to abstract...” “use manipulatives when investigating...”
 Differentiate Instruction	4.12.2 Collaboration provides kinesthetic, visual, and auditory entry points. Consider how to leverage this activity when selecting groupings.	
Lesson Notes:		

4.12.1 Warm-Up: Bell Work

Component Overview: Students write the equations of a line that is parallel and perpendicular to a given line through a point not on the line.



4.12.1 Warm-Up: Bell Work

- Write the equation of the line that goes through the point $(-27, 10)$ and is parallel to the line $y + 5 = \frac{5}{9}(x - 10)$.
 $y = \frac{5}{9}x + 25$ or any equivalent equation
- Write the equation of the line that goes through the point $(-27, 10)$ and is perpendicular to the line $y + 5 = \frac{5}{9}(x - 10)$.
 $y - 10 = -\frac{9}{5}(x + 27)$ or any equivalent equation



4.12.1 Warm-Up can be used as a formative assessment for measuring student learning. If your classroom uses different methods for formative assessments, infuse choice to increase student agency by allowing students to choose whether to answer question 1 or question 2. Their choice and work will be additional pieces of data for you as the class synthesizes their learnings.

4.12.2 Collaboration: Sorting Equations

Component Overview: Students work together to match equations based on three different features.



4.12.2 Collaboration: Sorting Equations

1. Work with your partner to sort the equations into pairs that have the given characteristic. Find three different sets of pairs of equations for each characteristic. Lines may be used more than once.

Answers may vary.

Pairs of Parallel Lines	Pairs of Perpendicular Lines	Pairs of Lines with the same y-intercept
<i>C & H</i>	<i>D & L</i>	<i>B & H</i>
<i>A & E</i>	<i>B & K</i>	<i>C & L</i>
<i>G & I</i>	<i>F & J</i>	<i>E & K</i>

A	$5x + 4y = 40$	G	$y + 25 = -\frac{5}{6}(x - 60)$
B	$y = 6x + 1$	H	$3x - y = -1$
C	$y - 11 = 3(x - 2)$	I	$y = -\frac{5}{6}x + 24$
D	$-3x + 2y = -12$	J	$y = \frac{5}{4}(x - 16)$
E	$y = -\frac{5}{4}x - 2$	K	$x + 6y = -12$
F	$4x + 5y = 120$	L	$y = -\frac{2}{3}x + 5$

2. Work with your partner to determine which pair of lines have the same x-intercept.

- A. A and G
- B. B and J
- C. D and K
- D. F and G**



Students apply their learning from Lessons 9-11 in this interactive, cumulative activity that incorporates all elements of parallel and perpendicular lines.

Consider using information from the previous three lessons to form a small group of students who are struggling working with equations of parallel and perpendicular lines. Facilitate these students as they work through the 4.12.2 Collaboration.



Provide kinesthetic and visual entry-points by having students move and create pairs of lines. If resources permit, consider having students at posters around the room constructing additional visual representations for matching pairs. Also provide an auditory entry-point by having students verbally process and share their findings.



The less information provided, the more complex the task will be. Avoid telling students how many pairs each equation has and avoid telling students that the third column only has two pairs. This will engage students in critical thinking.

4.12.2 Collaboration: Sorting Equations (continued)

3. Discuss with your partner which key features will be the same for any pair of parallel lines.

- ☐ decreasing/increasing
- ☐ domain
- ☐ range
- ☐ slope
- ☐ x -intercept
- ☐ y -intercept

4. Discuss with your partner which key features will be the same for any pair of perpendicular lines.

- ☐ decreasing/increasing
- ☐ domain
- ☐ range
- ☐ slope
- ☐ x -intercept
- ☐ y -intercept



Monitor while students are working to determine if there are any questions that need to be debriefed before moving to the next component. This could include highlighting student voices to share rich discussions or comments overhead while they collaborated for the whole class to hear or to address common misconceptions that were evident among most of the class.

If time allows or partners finish early, consider allowing students to use an only graphing utility such as the Desmos graphing calculator to verify their answers to questions 1-4.

They can work with their partners to resolve any errors or write/ask a question if they aren't sure how they made an error.

4.12.3 Your Turn!

Component Overview: Students write the equations of a line, in three different forms, that is parallel and perpendicular to a given line through a point not on the line.

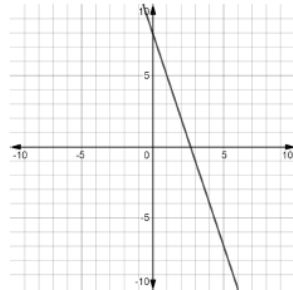


4.12.3 Your Turn!

- Write the equation of the line, in all three forms, that is perpendicular to the line $6x - 3y = 15$ and passes through the point $(1, -5)$.

Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 5 = -\frac{1}{2}(x - 1)$	$y = -\frac{1}{2}x - \frac{9}{2}$ or any equivalent equation in slope-intercept form	$\frac{1}{2}x + y = -\frac{9}{2}$ or any equivalent equation in standard form

- Write the equation of the line, in all three forms, that is parallel to the line shown on the coordinate grid and that passes through the point $(-4, -7)$.



Point-Slope Form	Slope-Intercept Form	Standard Form
$y + 7 = -3(x + 4)$	$y = -3x - 19$	$3x + y = -19$

- Write the equation of the line that passes through $(5, 9)$ and is parallel to the line that passes through the points $(-4, -1)$ and $(-1, -1)$.
 $y = 9$
- Write the equation of the line that passes through $(-12, 15)$ and is perpendicular to the line that passes through the points $(-2, -4)$ and $(3, 6)$.
 $y - 15 = -\frac{1}{2}(x + 12)$ or any equivalent equation



Notice that students are continuing to only write the equations of lines rather than graphing. For students who may benefit from visually checking their work, consider offering Desmos or GeoGebra as a digital tool. By graphing the lines in their different forms, students can verify that the lines are parallel.

Note: The benchmark targets writing, not graphing. Use graphing as a scaffolding tool to an end.



What do you notice about the coordinate pairs in question 3? How does that help you infer what the slope is before calculating it? How does this help you determine what the equation of the line will look? (This line of questioning helps target misconceptions of vertical and horizontal lines).

4.12.4 Wrap-Up: Error Analysis

Component Overview: Consider using all or parts of this exercise as a formative assessment to gather data on student progress from this lesson.



4.12.4 Wrap-Up: Error Analysis

- The work of two students is shown in the table. The students were asked to write the equation of the line that passes through $(-5, -9)$ and is perpendicular to the line that passes through the points $(3, -4)$ and $(0, 5)$. Find each person's error and correct it.

Sandip	Jiang
Slope = $\frac{\text{change in } y}{\text{change in } x} = \frac{-4-5}{3-0} = -3$	Slope = $\frac{\text{change in } y}{\text{change in } x} = \frac{-4-5}{0-3} = 3$
Equation: $y + 9 = -3(x + 5)$	Equation: $y + 9 = -\frac{1}{3}(x + 5)$

Sandip used the wrong slope in his equation. He used -3 for the slope when it should be $\frac{1}{3}$ because he is writing the equation of a perpendicular line, not a parallel line.

Jiang calculated the slope incorrectly. He did not calculate the change in x using the points in the same order as he calculated the change in y . The slope of the original line should be -3 so that the slope in the equation of the new line is $\frac{1}{3}$.

- What is the equation of the line that passes through $(-5, -9)$ and is perpendicular to the line that passes through the points $(3, -4)$ and $(0, 5)$?
 $y + 9 = \frac{1}{3}(x + 5)$ or any equivalent equation



MA.K12.MTR.4.1: Discussion and Reflection
 While students are not discussing in 4.12.4 Wrap-Up, they are applying the tools and skills of analyzing the work of others (error analysis).